

4(a)(i)	C = red blood cell / erythrocyte A red blood corpuscle D = lymphocyte A plasma cell R if also stated as granulocyte A B- or T-Cell I white blood cell / leucocyte / agranulocyte E = monocyte R if also stated as granulocyte I white blood cell / leucocyte / agranulocyte A macrophage F = neutrophil A eosinophil A basophil R if also stated as agranulocyte I white blood cell / leucocyte / granulocyte C correct = one mark D correct = one mark E and F correct = one mark	3
4(a)(ii)	<i>any two from:</i> <i>more white blood cells because</i> result of immune response ; clonal expansion / described, of, B- / T-, lymphocytes / cells ; idea of producing (large quantity of specific) antibody in context of immune response increased, production of macrophages / phagocytes ; <i>if written in context of HIV/AIDS as the infectious disease</i> fewer white blood cells ; further detail ; e.g. T-helper lymphocytes destroyed	2

Question	Answer	Marks
4(b)	1 decreased / 13% less, percentage oxygen saturation / percentage saturation haemoglobin with oxygen <u>and</u> similar (quantity) / same (quantity) / 0.1 cm³ per 100 cm³ blood less, oxygen delivered by haemoglobin to tissues ; <i>plus any three from:</i> 2 at high altitude lower partial pressure of oxygen, in atmosphere / in inhaled air / in alveolus / into pulmonary capillaries / AW ; 3 in lungs / in pulmonary capillaries / at the alveolus, less oxygen, binds to / taken up by / AW, haemoglobin ; 4 haemoglobin has lower affinity for oxygen (in lungs) ; 5 increased production of haemoglobin ; 6 increased, production / number / density, of red blood cells ; A increased haematocrit 7 (so) more haemoglobin available to bind oxygen / same quantity of oxygen can be taken up as at sea level ; 8 AVP ; e.g. ref. to decreased plasma volume increase in erythropoietin (EPO) concentration other described physiological change	4
4(c)	absence of, good health / well-being / lack of <u>physical</u> well-being / AW ; A disorder / illness further detail ; e.g. having an adverse effect reduced / impaired, function A described e.g. less oxygen, taken up / transported produces, (specific) signs / symptoms ; A described symptoms	2